



Brazilian copper mine bacteria

A bacterium can transform waste from copper mining into a form of pure copper.
<http://digit.in/may21-25>



Climate change and reproduction

A new study finds that climate change is affecting reproductive decision-making.
<http://digit.in/may21-26>

have tried to explain away this discrepancy by coming up with various explanations, such as all continental plates having accelerated movement during that time period, or thinning of the continental crust of India as it moved over hotspots of magma activity, that gave rise to the Rajmahal and Deccan traps, the remnants of massive ancient lava flows. Alternatively, the computer models using known constraints place the continents thousands of kilometers apart at the time of the collision.

Plate tectonics is actually a recent scientific revolution, and till the middle of the twentieth century, most scientists believed that the continents were more or less fixed. When geologists started piecing the puzzle together, they noticed bits and pieces of ancient continents wedged up in between continental plates, surrounded by the fault lines. These bits and pieces are known as terranes, and in the western Himalayas, wedged between Ladakh in India and Afghanistan, is the Kohistan terrane. The rocks in this region came from an ancient chain of volcanic islands in the Tethys Ocean, known as the Kohistan-Ladakh arc. The islands are similar to the Marina Islands in the Pacific Ocean today, and were volcanic in origin. The rocks in the Himalayas are proof that the Tethys Ocean was not empty, and that these islands were squished up into the Himalayas when India collided with Eurasia. Now the collision between the Indian plate and the Eurasian plate was no longer a simple story, because there were additional island chains or microcontinents in the Tethys Ocean. Since the mid 90s, researchers have used a wide variety of geological studies, ranging from stratigraphy to paleomagnetism, combined with computer modelling, to propose a number of theories on the formation of the Himalayas. In some of these models, the collision between the Indian plate and the Kohistan-Ladakh arc happened first, while in others, the Kohistan-Ladakh

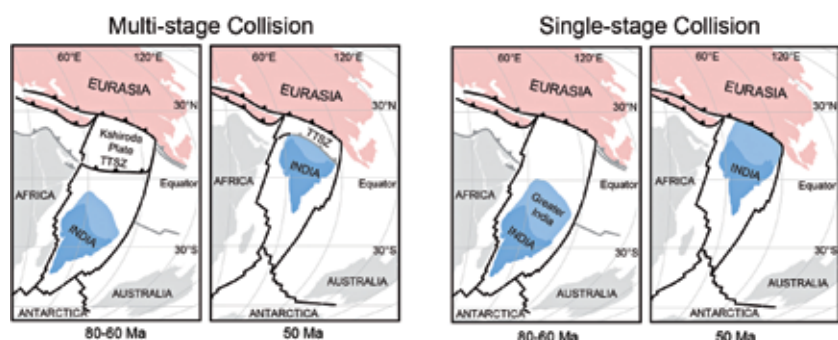
arc collided with Eurasia first, after which the Indian plate crashed into the ongoing collision; the debate continues to this day.

Now researchers from MIT have come up with a new study that sheds light on the problem. According to the study, there was another plate between the Indian plate and the Eurasian plate, the remnants of which is the Kohistan terrane. The researchers call this plate the Kshiroda plate, and propose a second subduction zone in the middle of the Tethys Ocean, called the Trans-Tethyan subduction zone, that also ran east to west. The team visited Ladakh in India, and gathered cork sized rock samples, noting their precise

location and orientation within the rock. Paleomagnetic studies of these samples allowed the team to put together the complete picture.

collided with Eurasia. During this time, the force of the collision caused the continents to fold and push rocks upwards, forming the features observed in the Himalayan rocks today. The presence of the Kshiroda plate changes the estimated size of the Indian plate before the collision, making it smaller and making it fit more neatly in the computer models.

Craig Martin, lead author of the study says, "I think it is cool to imagine idyllic tropical volcanic islands, with dinosaurs roaming around on them, having been sandwiched between two colliding tectonic plates and uplifted to form the roof of the world." It is hard to imagine the Himalayas being tropical islands at one point in the geological



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Around 80 million years ago, there were two subduction events simultaneously. The Kshiroda plate sank beneath the Eurasian plate, while at the same time, just south of the Kshiroda plate was the Trans-Tethyan subduction zone, where India collided with the Kohistan-Ladakh arc. Together, these two subduction zones ended up closing the Tethys ocean. The Kohistan-Ladakh arc acted to check the speed of the northward moving Indian plate. The two subduction zones were active till around 55 million years ago. Then, around 45 to 40 million years ago, around 10 million years after the generally accepted date, India

past, but in 2007, researchers found palm fossils in the area. The double subduction zone also explains another scientific mystery, the Earth began to cool around 66 million years ago, known as the Cenozoic global cooling. There have been many explanations for why it occurred. According to the new paper, between 80 to 66 million years ago the twin subduction zones in the Tethys Ocean, combined with the process of weathering and erosion, helped to trap and sequester carbon into the Earth, leading to a cooling of global climate in the Cenozoic era, ushering in a transition from a greenhouse world to an icehouse planet. The formation of the Himalayas has been identified as one of, if not the primary force behind the Cenozoic global cooling, that continues to this day.

Credit: Craig Martin et al, PNAS